



In addition to Part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course

Course No. : PHA G543
Course Title : CLINICAL RESEARCH
Instructor-in-charge: HEMANT R. JADHAV

1. Course Description:

Overview of new drug research and development; Bioethics and institutional review board; regulatory control of clinical trials for NDA & ANDA application; Good Clinical Practice (GCP); related ICH guidelines; applied clinical epidemiology and biostatistics; clinical trial study design; trial development (protocol, case report form, and data management); clinical trial management- coordinating clinical trial at the site, documentation methodology, implementing monitoring plan and performing quality control; statistical analysis and data interpretation; monitoring obligations and methods; and Medical writing & report preparation for various submissions.

2. Scope and Objective of the course:

There is a growing need for clinical researchers in the health industries (pharmaceutical, biotechnology/medical device companies, research institutes, and hospitals) involved in the development of new drugs and therapies. This course is aimed at training students in conducting clinical trials on humans with new drugs/therapies before they are introduced to the market. Students completing the coursework will gain specialized knowledge and skills required to design, monitor, and manage clinical trials. Courses include the drug development process; regulations, guidelines, and standards; research methodology and biostatistics; clinical trial organization, monitoring, and documentation; and project management.

3. Text Book (T):

- T1: Lawrence M Friedman, Curt D Furberg and David L DeMets, "Fundamentals of Clinical Trials", Springer Verlag, New York, 5th Ed., 2015.
T2: Shein-Chung Chow, Jen-Pei Liu, "Design and Analysis of Clinical Trials", Wiley-IEEE, 2013.

Reference Books (R):

- R1: Steven Piantadosi, "Clinical Trials-A Methodologic Perspective", John Wiley & Sons, 2005.
R2: R A Guarino, "New Drug Approval Process" Marcel Dekker, New York, 2nd Edn., Vol. 56, 1993.



4. Course Plan:

Sr.	Topic to be covered	Reference	Learning outcomes
1.	Introduction to clinical research	T1 Ch. 1	Overview of clinical trials
2.	Regulatory control of clinical trials for NDA & ANDA applications	R2 Ch. 1-3, 5, 10, 11	Understanding US-FDA requirements for clinical trials
3.	Decision on goals; primary questions to be answered; decision on response variables, population, and patient selection	T1 Ch. 2 & 3 T2 Ch. 3	Basic design considerations for clinical research
4.	Designs for clinical trials, classification of clinical trials, blinding, randomization techniques, baseline assessment	T1 Ch. 4, 5, 6 & 8 T2 Ch. 3, 4, 7	Understanding methods to reduce bias in clinical research
5.	Clinical trial protocol, recruitment, case report form, and data management	T1 Ch. 9 T2 Ch. 14	Clinical trial protocol-related documentation
6.	Data quality and its control, problems in data collection, and methods to minimize poor data, participant compliance, monitoring response variables, and study termination methods	T1 Ch. 10, 13, 15, 17 T2 Ch. 8, 12	Clinical research data management

5. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Remarks
Mid-semester Test	90 min	30	9/10 FN2	CB
Continuous assessment		35	Continuous	
Comprehensive Exam	120 min	35	8/12 AN	CB and OB

*Continuous assessment will be based on the theory covered in the class. Topics and numbers will be announced in class. It will be in terms of home assignments, tutorials, projects, laboratory, viva voce, and class participation

Reading Assignments: Students are advised to read and collect additional information on the above-mentioned topics from journals and other online sources.

Attendance: Although attendance is not compulsory, regularity in theory and practical classes will be a decisive factor during grading, especially in borderline cases.

Chamber Consultation Hour: To be announced in the class.

Make-up policy: Generally, make-up will be considered for regular students only.

Notices: Notices concerning this course will be displayed on Pharmacy N. B.

**Instructor-in-Charge
PHA G543**